

# Moveit Trajectory Planning

---

## Moveit Trajectory Planning

1. Usage Environment
2. Driving the Real Machine
  - 2.1 Starting the Real Machine
  - 2.2 Starting MoveIt2
3. Trajectory Planning
  - 3.1. Visualization
  - 3.2. Launch Command

## 1. Usage Environment

---

Ubuntu 22.04

ROS2: Humble

## 2. Driving the Real Machine

---

Driving the real machine involves subscribing to Moveit2's `/joint_states` topic to convert the joint state information of the robotic arm into control data for the actual robotic arm.

**Note: Since the real robotic arm lacks obstacle avoidance, some poses may encounter obstacles; therefore, the planned robotic arm movements should be as reasonable as possible and avoid locations with obstacles.**

**(It is recommended to use preset poses for demonstrating the real machine.)**

### 2.1 Starting the Real Machine

Without driving the real machine, the robotic arm movements will be simulated and demonstrated in MoveIt:

```
ros2 run dofbot_driver dofbot_driver
```

### 2.2 Starting MoveIt2

```
ros2 launch dofbot_moveit demo.launch.py
```

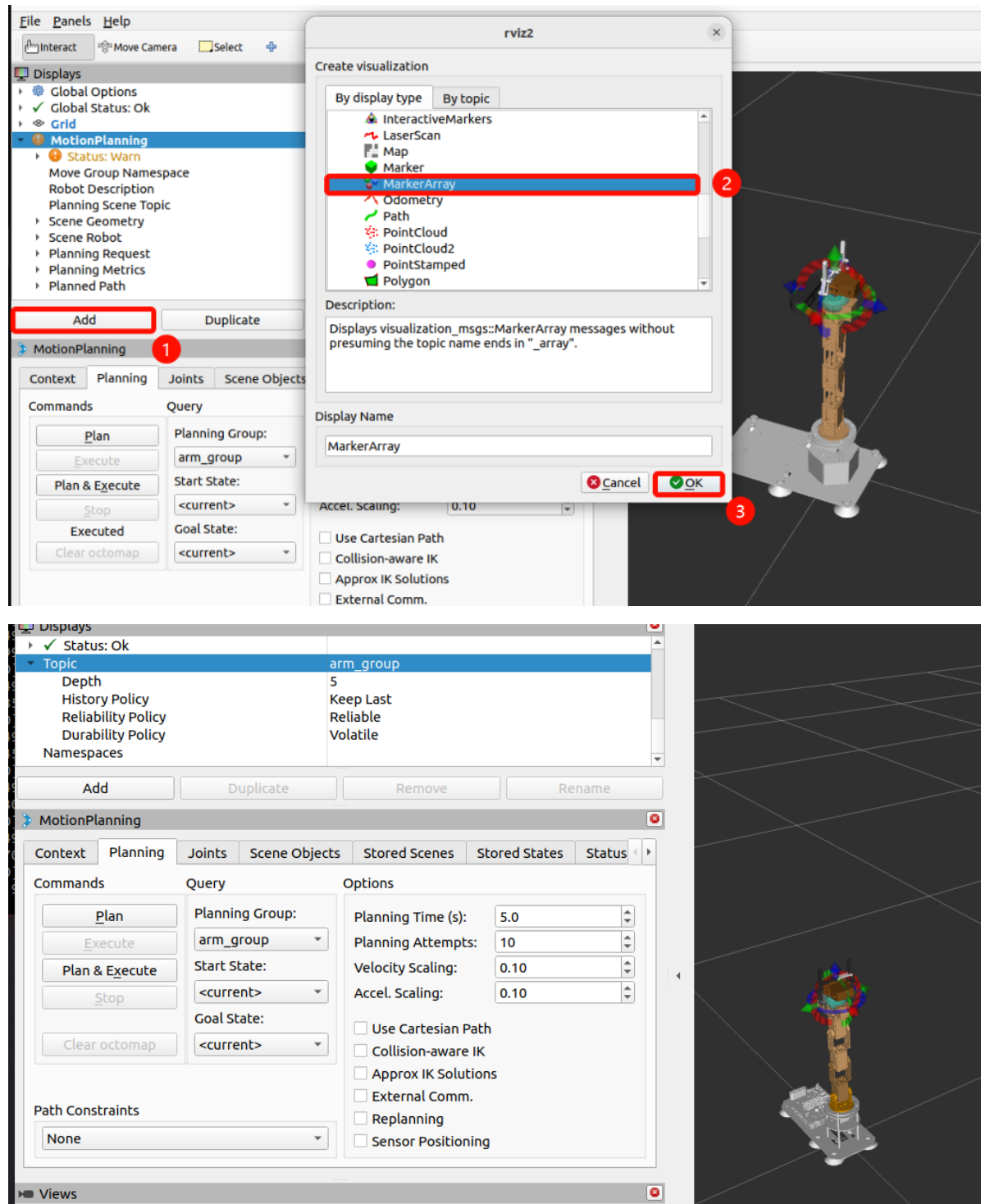
## 3. Trajectory Planning

---

After the program runs, the robotic arm will plan the trajectory to the target position, and the robotic arm's end effector will draw the motion trajectory in RViz2.

### 3.1. Visualization

Before executing the launch command, you must add the `MarkerArray` plugin in `RViz2` to display the planned path. For the `MarkerArray`, you need to select the `arm_group` topic.



## 3.2. Launch Command

Once the robotic arm has successfully loaded in MoveIt and the message `You can start planning now!` appears, run the following command: the robotic arm will automatically plot the trajectory of the planned path.

```
yahboom@yahboom-virtual-machine: ~ 80x24
PLPlanner
[move_group-3] [INFO] [1773993244.096183272] [moveit_move_group_capabilities_base.move_group_context]: MoveGroup context initialization complete
[move_group-3] Loading 'move_group/ApplyPlanningSceneService'...
[move_group-3] Loading 'move_group/ClearOctomapService'...
[move_group-3] Loading 'move_group/MoveGroupCartesianPathService'...
[move_group-3] Loading 'move_group/MoveGroupExecuteTrajectoryAction'...
[move_group-3] Loading 'move_group/MoveGroupGetPlanningSceneService'...
[move_group-3] Loading 'move_group/MoveGroupKinematicsService'...
[move_group-3] Loading 'move_group/MoveGroupMoveAction'...
[move_group-3] Loading 'move_group/MoveGroupPlanService'...
[move_group-3] Loading 'move_group/MoveGroupQueryPlannersService'...
[move_group-3] Loading 'move_group/MoveGroupStateValidationService'...
[move_group-3]
[move_group-3] You can start planning now!
[move_group-3]
[rviz2-4] [INFO] [1773993244.191243764] [move_group_interface]: Ready to take commands for planning group arm_group.
[rviz2-4] [INFO] [1773993244.192377244] [moveit_ros_visualization.motion_planning_frame]: group arm_group
[rviz2-4] [INFO] [1773993902.039456161] [moveit_ros_visualization.motion_planning_frame_objects]: Loaded scene geometry from '/home/yahboom/dofbot_ws/src/dofbot_moveit/scene/shape.scene'
```

```
ros2 run dofbot_moveit multi_track_motion
```

